

Harsh Agarwal

Jaipur, Rajasthan

+91 8209305539 | harshagarwal819@gmail.com

linkedin.com/in/harshaga819 | github.com/harshaga819 | harshhub.netlify.app/

PROFILE

Computer Science graduate with hands-on experience in Machine Learning, Deep Learning, and Generative AI. Experienced in building end-to-end AI applications using Python, LangChain, and RAG. Strong foundation in LLMs, Semantic Search, Vector Databases, and modern AI workflows.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Arya College of Engineering and IT, Jaipur | 2022 – 2026

CGPA - **8.31**

Senior Secondary (Class XII), CBSE

MPS International School, Jaipur | 2021 – 2022

Percentage - **80.5%**

TECHNICAL SKILLS

Languages: Python, C++

Database: MySQL

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow

Generative AI: RAG, Vector Databases, Semantic Search, Embeddings, Document Retrieval

Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation

Deep Learning: ANN, CNN, RNN, LSTM, GRU

PROJECTS

PDF Chatbot — RAG-based AI Assistant

Tech Stack: Python, LangChain, ChromaDB, Hugging Face, Sentence Transformers, Streamlit

<https://github.com/harshaga819/PDFChatbot.git>

- Built a production-style PDF Chatbot using **LangChain**, **Hugging Face embeddings**.
- Implemented **document ingestion**, **chunking**, **vector embeddings**, and **semantic search**.
- Enabled accurate context-aware question answering through a **Retrieval-Augmented Generation (RAG) pipeline**.
- Reduced document retrieval latency through optimized chunking and embedding pipeline.

Facial Emotion Recognition System — Deep Learning-Based Emotion Classification

Tech Stack: TensorFlow, Keras, OpenCV, MobileNetV2, transfer learning

<https://github.com/harshaga819/Facial-Emotion-Recognition-System.git>

- Developed a **deep learning**-based facial emotion recognition system using **transfer learning**.
- Applied image **preprocessing**, **data augmentation** on the CK+ dataset.
- Implemented **MobileNetV2 architecture** for multi-class emotion classification.
- Achieved approximately **80%** validation accuracy across **7** emotion classification.

CERTIFICATES

SQL (Intermediate) - Hacker-Rank

Problem Solving (Intermediate) - Hacker-Rank

Introduction to Generative AI - SkillUP

ACHIEVEMENTS

Solved **150+** DSA problems on LeetCode and HackerRank

Earned **4** Star in C++ and **3** Star in SQL on HackerRank